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BATTERY INCLUDING SEPARATOR AND ELECTRONIC DEVICE INCLUDING SAME

Abstract

A battery is provided. The battery includes a first electrode, a second electrode spaced apart from the first electrode, and a separator including a fiber disposed between the first electrode and the second electrode, the fiber including a core and a sheath at least partially surrounding the core, wherein the core of the fiber includes a phase change material, and wherein the sheath of the fiber includes a polymer material.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation application, claiming priority under 35 U.S.C. § 365(c), of an International application No. PCT/KR2023/015273, filed on Oct. 4, 2023, which is based on and claims the benefit of a Korean patent application number 10-2022-0152860, filed on Nov. 15, 2022, in the Korean Intellectual Property Office, and of a Korean patent application number 10-2022-0175719, filed on Dec. 15, 2022, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

[0002] The disclosure relates to a battery including a separator and an electronic device including the same.

2. Description of Related Art

[0003] An electronic device may include various electronic components. The electronic device may include a battery for providing power to the electronic components. The battery may include a separator disposed between a positive electrode and a negative electrode. The separator may include pores for passing ions (e.g., lithium ions).

[0004] The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

SUMMARY

[0005] Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide a battery including a separator and an electronic device including the same.

[0006] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

[0007] In accordance with an aspect of the disclosure, a battery is provided. The battery includes a first electrode, a second electrode spaced apart from the first electrode, and a separator including a fiber disposed between the first electrode and the second electrode, the fiber including a core and a sheath at least partially surrounding the core, wherein the core of the fiber includes a phase change material, and wherein the sheath of the fiber includes a polymer material.

[0008] In accordance with another aspect of the disclosure, an electronic device is provided. The electronic device includes a battery configured to provide power to at least one component of the electronic device, and power management integrated circuitry (PMIC) configured to manage the power provided from the battery to the at least one component, wherein the battery includes a first electrode, a second electrode spaced apart from the first electrode, and a separator including a fiber disposed between the first electrode and the second electrode, the fiber including a core and a sheath at least partially surrounding the core, wherein the core of the fiber includes a phase change material, and wherein the sheath of the fiber includes a polymer material.

[0009] Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0011] FIG. 1 is a block diagram of an electronic device in a network environment according to an embodiment of the disclosure;

[0012] FIG. 2A is a block diagram of a power management module and a battery according to an embodiment of the disclosure;

[0013] FIG. 2B is a block diagram of an electronic device according to an embodiment of the disclosure;

[0014] FIG. 3A schematically illustrates a battery according to an embodiment of the disclosure;

[0015] FIG. 3B is an enlarged view of an X region of the battery of FIG. 3A according to an embodiment of the disclosure;

[0016] FIG. 3C is a schematic exploded perspective view of a battery according to an embodiment of the disclosure;

[0017] FIG. 4A schematically illustrates an electrospinning device for manufacturing a separator according to an embodiment of the disclosure;

[0018] FIG. 4B schematically illustrates a separator manufactured by the manufacturing method of FIG. 4A according to an embodiment of the disclosure;

[0019] FIG. 5A is a cross-sectional view of a separator cut along line A-A' of FIG. 4B according to an embodiment of the disclosure;

[0020] FIG. 5B is a cross-sectional view of a separator cut along line B-B' of FIG. 4B according to an embodiment of the disclosure;

[0021] FIG. 6A schematically illustrates a state in which a separator absorbs heat energy according to an embodiment of the disclosure;

[0022] FIG. 6B schematically illustrates a separator that has absorbed heat energy according to an embodiment of the disclosure; and

[0023] FIG. 7 illustrates a state in which an external object penetrates a battery according to an embodiment of the disclosure.

[0024] The same reference numerals are used to represent the same elements throughout the drawings.

DETAILED DESCRIPTION

[0025] The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

[0026] The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

[0027] It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

[0028] It should be appreciated that the blocks in each flowchart and combinations of the

flowcharts may be performed by one or more computer programs which include instructions. The entirety of the one or more computer programs may be stored in a single memory device or the one or more computer programs may be divided with different portions stored in different multiple memory devices.

[0029] Any of the functions or operations described herein can be processed by one processor or a combination of processors. The one processor or the combination of processors is circuitry performing processing and includes circuitry like an application processor (AP, e.g. a central processing unit (CPU)), a communication processor (CP, e.g., a modem), a graphics processing unit (GPU), a neural processing unit (NPU) (e.g., an artificial intelligence (AI) chip), a Wi-Fi™ chip, a Bluetooth™ chip, a global positioning system (GPS) chip, a near field communication (NFC) chip, connectivity chips, a sensor controller, a touch controller, a finger-print sensor controller, a display driver integrated circuit (IC), an audio CODEC chip, a universal serial bus (USB) controller, a camera controller, an image processing IC, a microprocessor unit (MPU), a system on chip (SoC), an IC, or the like.

[0030] FIG. 1 is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure.

[0031] Referring to FIG. 1, an electronic device **101** in a network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

[0032] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to an embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0033] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main

processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0034] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0035] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0036] The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0037] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0038] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0039] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0040] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0041] The interface **177** may support one or more specified protocols to be used for the electronic

device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0042] The connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, an HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphone connector).

[0043] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0044] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0045] The power management module **188** may manage power supplied to the electronic device **101**. According to an embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0046] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0047] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi™) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a fifth-generation (5G) network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0048] The wireless communication module **192** may support a 5G network, after a fourth-generation (4G) network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module

192 may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 gigabits per second (Gbps) or more) for implementing eMBB, loss coverage (e.g., 164 decibels (dB) or less) for implementing mMTC, or U-plane latency (e.g., 0.5 milliseconds (ms) or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

[0049] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

[0050] According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, an RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0051] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0052] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices (e.g., the electronic devices **102** and **104** and the server **108**). For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-

latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0053] FIG. 2A is a block diagram of a power management module and a battery according to an embodiment of the disclosure.

[0054] FIG. 2B is a block diagram of an electronic device according to an embodiment of the disclosure.

[0055] FIG. 2A is a block diagram illustrating a power management module and a battery according to an embodiment of the disclosure.

[0056] Referring to FIG. 2A, block diagram **200** illustrates that a power management module **188** may include charging circuitry **210**, a power adjuster **220**, or a power gauge **230**. The charging circuitry **210** may charge a battery **189** by using power supplied from an external power source outside the electronic device **101**. According to an embodiment, the charging circuitry **210** may select a charging scheme (e.g., normal charging or quick charging) based at least in part on a type of the external power source (e.g., a power outlet, a USB, or wireless charging), magnitude of power suppliable from the external power source (e.g., about 20 Watt or more), or an attribute of the battery **189**, and may charge the battery **189** using the selected charging scheme. The external power source may be connected with the electronic device **101**, for example, directly via the connecting terminal **178** or wirelessly via the antenna module **197**.

[0057] The power adjuster **220** may generate a plurality of powers having different voltage levels or different current levels by adjusting a voltage level or a current level of the power supplied from the external power source or the battery **189**. The power adjuster **220** may adjust the voltage level or the current level of the power supplied from the external power source or the battery **189** into a different voltage level or current level appropriate for each of some of the components included in the electronic device **101**. According to an embodiment, the power adjuster **220** may be implemented in the form of a low drop out (LDO) regulator or a switching regulator. The power gauge **230** may measure use state information about the battery **189** (e.g., a capacity, a number of times of charging or discharging, a voltage, or a temperature of the battery **189**).

[0058] The power management module **188** may determine, using, for example, the charging circuitry **210**, the power adjuster **220**, or the power gauge **230**, charging state information (e.g., lifetime, over voltage, low voltage, over current, over charge, over discharge, overheat, short, or swelling) related to the charging of the battery **189** based at least in part on the measured use state information about the battery **189**. The power management module **188** may determine whether the state of the battery **189** is normal or abnormal based at least in part on the determined charging state information. If the state of the battery **189** is determined to abnormal, the power management module **188** may adjust the charging of the battery **189** (e.g., reduce the charging current or voltage, or stop the charging). According to an embodiment, at least some of the functions of the power management module **188** may be performed by an external control device (e.g., the processor **120**).

[0059] The battery **189**, according to an embodiment, may include a protection circuit module (PCM) **240**. The PCM **240** may perform one or more of various functions (e.g., a pre-cutoff function) to prevent a performance deterioration of, or a damage to, the battery **189**. The PCM **240**, additionally or alternatively, may be configured as at least part of a battery management system (BMS) capable of performing various functions including cell balancing, measurement of battery capacity, count of a number of charging or discharging, measurement of temperature, or measurement of voltage.

[0060] According to an embodiment, at least part of the charging state information or use state

information regarding the battery **189** may be measured using a corresponding sensor (e.g., a temperature sensor) of the sensor module **176**, the power gauge **230**, or the power management module **188**. According to an embodiment, the corresponding sensor (e.g., a temperature sensor) of the sensor module **176** may be included as part of the PCM **240**, or may be disposed near the battery **189** as a separate device.

[0061] Referring to FIG. 2B, block diagram **300** illustrates that an electronic device **101** may include a battery **189** and power management integrated circuitry (PMIC) **360** (e.g., the power management module **188** of FIG. 2A). According to an embodiment, the electronic device **101** may include components (e.g., a camera, a printed circuit board) for various functions. According to an embodiment, the battery **189** used in the electronic device **101** may be configured to provide power to components of the electronic device **101**. The electronic device **101** may include the PMIC **360** to manage power provided from the battery **189** to the components. The PMIC **360** may be configured to convert power provided from the battery **189** into power required by each of the components and distribute the converted power to each of the components.

[0062] FIG. 3A schematically illustrates a battery according to an embodiment of the disclosure.

[0063] FIG. 3B is an enlarged view of an X region of the battery of FIG. 3A according to an embodiment of the disclosure. FIG. 3C is a schematic exploded perspective view of a battery according to an embodiment of the disclosure.

[0064] Referring to FIGS. 3A and 3B, a battery **189** according to an embodiment may include a first electrode **310**, a second electrode **320**, and a separator **330**. The battery **189** may include an electrolyte **350** that enables movement of ions (e.g., lithium cations) between the first electrode **310** and the second electrode **320**, and/or a case **340** that forms an exterior of the battery **189**. The first electrode **310**, the second electrode **320**, and the separator **330** may be accommodated within the case **340**. The electrolyte **350** may enable movement of ions for an electrochemical reaction of the first electrode **310** and the second electrode **320**. For example, the separator **330** may include pores (e.g., pores **330a** of FIG. 3B) through which ions can pass. The case **340** may accommodate the first electrode **310**, the second electrode **320**, the separator **330**, and the electrolyte **350**. According to a shape of the case **340**, a type of the battery **189** may be distinguished. For example, as illustrated in FIG. 3A, the battery **189** may be a roll-type battery in which components of the battery **189** are rolled within a cylindrical case, but is not limited thereto. For example, the battery **189** may include a square, cylindrical, or pouch-shaped case **340**. The battery **189** may be a stack-type battery **189** in which components within the case **340** are alternately laminated.

[0065] According to an embodiment, the first electrode **310** and the second electrode **320** may be electrically different. For example, the first electrode **310** may be referred to as a positive electrode in which cations (e.g., lithium ions), during discharge, receive electrons and are reduced. The first electrode **310** may include a positive electrode substrate **310a** coated with a positive electrode active material **310b**. For example, the second electrode **320** may be referred to as a negative electrode in which cations (e.g., lithium ions), during discharge, release electrons and are oxidized. The second electrode **320** may include a negative electrode substrate **320a** coated with a negative electrode active material **320b**. According to an embodiment, the first electrode **310** and the second electrode **320** may be spaced apart from each other. For example, the second electrode **320** may be spaced apart from the first electrode **310**.

[0066] According to an embodiment, the separator **330** may be disposed between the first electrode **310** and the second electrode **320** so that the first electrode **310** and the second electrode **320** do not contact each other. The separator **330** may be configured to prevent a short circuit due to contact between the first electrode **310** and the second electrode **320**, by physically separating the first electrode **310** and the second electrode **320**. For example, when the first electrode **310** and the second electrode **320** are in direct physical contact, a short circuit current may flow along the first electrode **310** and the second electrode **320**. Since the short circuit current may generate local heat inside the battery **189**, a temperature of the battery **189** may rapidly increase, which may cause a

fire in the battery **189**.

[0067] According to an embodiment, the separator **330** may be configured to pass ions of the electrolyte **350**. For example, the separator **330** may be a porous separator **330** including fine pores **330a** for passing ions. For example, the separator **330** may include fine pores **330a** formed between fibers **331**. For example, lithium ions in the electrolyte **350** may be movable between the first electrode **310** and the second electrode **320** through the pores **330a** of the separator **330**.

[0068] Referring to FIGS. **3B** and **3C**, a separator **330** may include a fiber having a core-sheath structure (e.g., fiber **331** of FIG. **3C**). According to an embodiment, the separator **330** may be composed of a fiber **331** having a core-sheath structure, which is manufactured by electrospinning method. The separator **330** may be composed of a web formed by the fiber **331** having the core-sheath structure. The electrospinning may apply a high voltage to a solution having viscosity and reduce surface tension of the solution by utilizing a repulsive force of the solution formed by the high voltage. The solution with reduced surface tension may be discharged in a jet form to form the fiber **331**. In the separator **330** manufactured by the electrospinning, a core **332** may be disposed in the center. The separator **330** may include the fiber **331** including a sheath **333** that at least partially surrounds the core **332**. A method for manufacturing the separator **330** by the electrospinning according to an embodiment will be exemplified with reference to FIGS. **6A** and **6B**.

[0069] According to an embodiment, the separator **330** may be a porous separator. For example, the fiber **331** having a core-sheath structure may be formed by electrospinning. The fiber **331** of the core-sheath structure may be arranged in a grid pattern to form the separator **330**. The fiber **331** forming the separator **330** may include a phase change material in the core **332**. The fiber **331** forming the separator **330** may include a polymer material within the sheath **333**. A space between fibers **331** forming the separator **330** may be pores **330a** through which the electrolyte **350** passes. A porosity of the separator **330** may be about 30% to 80%, but is not limited thereto. For example, the porosity of the separator **330** may indicate a volume ratio of the pores **330a** with respect to a total volume of the separator **330**.

[0070] According to an embodiment, the core **332** may include a phase change material. The phase change material may be a material capable of controlling the temperature of the battery **189** by storing a large amount of heat energy or releasing the stored heat energy via a phase change process. For example, when the temperature inside the battery **189** is changed from a temperature lower than a melting point of the phase change material to a temperature higher than the melting point, the phase change material may be changed from solid phase to liquid phase. When the phase change material is changed from solid phase to liquid phase, the phase change material may absorb heat energy. According to an embodiment, the phase change material included in the core **332** may include at least one of paraffin, polyethylene glycol, sodium acetate trihydrate, sodium hydroxide monohydrate, magnesium nitrate hexahydrate, myristic acid, stearic acid, and xylitol. However, the phase change materials described above are not limited thereto. For example, the phase change material may include an inorganic hydrate (e.g., $\text{Na.sub.2HPO.sub.4.Math.12H.sub.2O}$, $\text{Na.sub.2SO.sub.4.Math.10H.sub.2O}$, or $\text{Zn(NO.sub.3).Math.6H.sub.2O}$), an inorganic salt, and/or a salt hydrate.

[0071] According to an embodiment, the sheath **333** may include a polymer material. According to an embodiment, the polymer material may include at least one of polyvinylidene fluoride (PVDF), polydimethylsiloxane (PDMS), polyethylene (PE), polyvinyl alcohol (PVA), and polyvinyl pyrrolidone (PVP). However, the polymer materials described above are merely examples, and are not limited thereto.

[0072] According to an embodiment, since the sheath **333** is disposed to at least partially surround the core **332**, the sheath **333** may form at least a portion of an outer surface of the fiber **331**. In order to prevent foreign substances inside the battery **189** from passing through the separator **330**, the separator **330** may be composed of a material having high mechanical strength. Since the sheath **333** forming the outer surface of the fiber **331** includes a polymer material, the mechanical strength

of the separator **330** may be determined by the polymer material. The polymer material may include a metallic oxide to provide high mechanical strength to the separator **330**. For example, the polymer material may include Al_2O_3 , AlOOH , or $\text{Mg}(\text{OH})_2$.

[0073] According to an embodiment, the separator **330** including the fiber **331** having the core-sheath structure may reduce a temperature increase of the battery **189**. According to an embodiment, a phase change material in the core **332** may be changed from solid phase to liquid phase by absorbing ambient heat energy when a temperature of the battery **189** increases due to a short circuit. When the phase change material undergoes the phase change, the phase change material may absorb heat energy corresponding to melting enthalpy. The heat energy absorption due to the phase change of the phase change material may reduce the temperature increase of the battery **189**. The separator **330** may have a relatively thin thickness by forming a separation membrane through the core-sheath structure, without a separate coating layer. For example, the sheath **333** forming the outer surface of the separator **330** may be in contact with the electrolyte **350** within the case **340**.

[0074] According to an embodiment, the separator **330** including the fiber **331** having the core-sheath structure may be configured to compensate for a contraction of the separator **330** due to a temperature increase. When a temperature of the battery **189** increases, the separator **330** may contract as a size of the pores **330a** decreases. When the separator **330** contracts, an internal short circuit may occur due to contact between the first electrode **310** and the second electrode **320**. According to an embodiment, when the temperature increases, the core-sheath of the separator **330** may expand. The expansion may reduce the amount of contraction of the contracted portion of the separator **330** by compensating for the contraction of the separator **330**. According to an embodiment, the separator **330** may prevent an internal short circuit due to the contraction.

[0075] FIG. **4A** schematically illustrates an electrospinning device for manufacturing a separator according to an embodiment of the disclosure.

[0076] FIG. **4B** schematically illustrates a separator manufactured by the manufacturing method of FIG. **4A** according to an embodiment of the disclosure.

[0077] FIG. **5A** is a cross-sectional view of a separator cut along line A-A' of FIG. **4B** according to an embodiment of the disclosure.

[0078] FIG. **5B** is a cross-sectional view of a separator cut along line B-B' of FIG. **4B** according to an embodiment of the disclosure.

[0079] Referring to FIG. **4A**, a separator (e.g., the separator **330** of FIG. **3C**) may be manufactured by electrospinning. The electrospinning is a method of manufacturing a continuous fiber **331** having a width in a range of micrometers to nanometers by using an electric field. When manufacturing the separator **330** by electrospinning, the separator **330** may have a high porosity. The separator **330** may have a large surface area. A structure and a size of the separator **330** may be easily controlled.

[0080] According to an embodiment, the separator **330** may be manufactured by an electrospinning device **400**. For example, the electrospinning device **400** may include a first container **410** that provides a first solution in which a phase change material is dissolved in a solvent, and a second container **430** that provides a second solution in which a polymer material is dissolved in a solvent. The electrospinning device **400** may include a nozzle tip **450**, which is connected to a first duct **420** extending from the first container **410** and a second duct **440** extending from the second container **430**. The first solution and a second solution may be discharged through the nozzle tip **450**. When being discharged through the nozzle tip **450**, the first solution and the second solution may be discharged independently without being mixed.

[0081] According to an embodiment, flow rates of the first solution and the second solution may be controlled through a first pump connected to the first container **410** and a second pump connected to the second container **430**. A power supply **460** may apply a high voltage (e.g., about 30 kV) to the nozzle tip **450**. The first solution and the second solution may be charged by the power supply

460 while passing through the nozzle tip **450**. The solutions may form a jet at the nozzle tip **450** due to electrostatic repulsion between charges applied to the solutions and Coulomb force applied to an external electric field. The jet may be elongated into a conical shape (Taylor cone shape). As a solvent volatilizes while a jet of a molten material including a polymer material and a phase change material reaches a current collector (or current collecting screen) **470** disposed below the nozzle tip **450**, a fiber **331** having a core-sheath structure may be obtained on the current collector **470**. The phase change material and the polymer material may have properties that do not mix with each other to form the core-sheath structure.

[0082] Referring to FIG. 4B, a separator **330** manufactured by electrospinning may be porous. Ions (e.g., lithium ions) in an electrolyte (e.g., the electrolyte **350** of FIG. 3A) may pass through the separator **330** through pores **330a** between fibers **331** forming the separator **330**. However, it is not limited thereto. For example, the fibers **331** may be arranged in a grid pattern. For example, the pores **330a** may be formed between the fibers **331**. According to an embodiment, a porosity of the separator **330** may be about 30% to 80%, but is not limited thereto. When the separator **330** is manufactured by electrospinning, a separator **330** including a fiber **331** having a core-sheath structure may be manufactured. The fiber **331** forming the separator **330** may include a phase change material within a core **332**. The fiber **331** forming the separator **330** may include a polymer material within a sheath **333**.

[0083] Referring to FIGS. 5A and 5B, a separator (e.g., the separator **330** of FIG. 3C) may include a fiber **331** having a core-sheath. According to an embodiment, the separator **330** may be manufactured by electrospinning, but is not limited thereto. The core **332** may include a phase change material. The sheath **333** surrounding at least a portion of the core **332** may include a polymer material. When a temperature of a battery **189** increases, the phase change material in the core **332** may be changed from solid phase to liquid phase by absorbing heat energy of a system. Since the phase change material absorbs the heat energy of the system when changing phase, the temperature increase of the battery **189** may be reduced. The sheath **333** including a polymer material may reduce the outflow of phase change material that has changed into liquid phase from the core **332** by surrounding at least a portion of the core **332**.

[0084] According to an embodiment, a width w_2 of the core **332** may be about 50% or more of a width w_1 of the fiber **331**. For example, the width w_1 of the fiber **331** may be about 1 nanometer (nm) to about 1,000 nm. The width w_2 of the core **332** may be about 0.5 nm to about 500 nm or more. A width of the sheath **333**, excluding the width w_2 of the core **332** from the width w_1 of the fiber **331**, may be about 0.5 nm to about 500 nm, but is not limited thereto. For example, the width w_1 of the fiber **331** may be referred to as a nano fiber of about 100 nm or less, but is not limited thereto. The width w_1 of the fiber **331** may be changed based on design of the battery **189**.

According to an embodiment, the core **332** may include 10 weight percent (wt %) to 90 wt % of the phase change material based on the weight of the core **332**, but is not limited thereto. The phase change material may be 10 wt % to 90 wt % of the core **332**.

[0085] According to an embodiment, since the separator **330** includes a phase change material inside the fiber **331**, it may have a relatively thin thickness. For example, when a phase change material for controlling a temperature of the battery **189** is coated on a surface of the fiber **331**, a width of the fiber **331** constituting the separator **330** may increase by a thickness of the coating layer including the phase change material. According to an embodiment, the separator **330** may not include a separate coating layer including the phase change material because it has a core-sheath structure including the phase change material in the core **332** rather than a structure for coating the phase change material on the surface of the fiber **331**. When a thickness of the separator **330** is thick, the amount of a positive electrode active material and/or a negative electrode active material included in the battery **189** may decrease, so energy density of the battery **189** may decrease. According to an embodiment, the separator **330** may be a relatively thin because it does not include the coating layer including the phase change material. As the thickness of the separator **330**

becomes thinner, the amount of the positive electrode active material and/or the negative electrode active material included in the battery **189** may increase, so the battery **189** according to an embodiment may have a relatively high energy density.

[0086] According to an embodiment, for mechanical strength of the separator **330**, the sheath **333** may include a polymer material. The polymer material may provide the mechanical strength to the separator **330**. The polymer material may protect the core **332** surrounded by the sheath **333**.

According to an embodiment, the separator **330** may provide stability to the battery **189**, while controlling a temperature of the battery **189** without reducing the energy density of the battery **189** by having a thin thickness.

[0087] FIG. **6A** schematically illustrates a state in which a separator absorbs heat energy according to an embodiment of the disclosure.

[0088] FIG. **6B** schematically illustrates a separator that has absorbed heat energy according to an embodiment of the disclosure.

[0089] Referring to FIG. **6A**, a separator **330** may be disposed between a first electrode **310** and a second electrode **320**. The separator **330** may prevent occurrence of a short circuit due to contact between the first electrode **310** and the second electrode **320** by separating the first electrode **310** and the second electrode **320** from each other.

[0090] According to an embodiment, the separator **330** may be configured to reduce a temperature increase of a battery **189**. The temperature of the battery **189** may abnormally increase due to various causes. For example, a short circuit caused by contact between the first electrode **310** and the second electrode **320**, overcharge of the battery **189**, an impact from the outside of the battery **189**, and/or a malfunction of a battery protection circuit (e.g., the protection circuit module **240** of FIG. **2A**) may cause overheating of the battery **189**. When the battery **189** is continuously overheated, the battery **189** may catch fire.

[0091] According to an embodiment, the separator **330** may include a fiber **331** including a core-sheath. A core **332** may include a phase change material. A sheath **333** may include a polymer material. The phase change material included in the core **332** may be changed from solid phase to liquid phase by absorbing heat energy E. For example, a melting point of the phase change material may be about 28° C. to about 90° C., but is not limited thereto. For example, the phase change material may be in solid phase in a temperature below the melting point. When a temperature of a system increase above a melting point temperature of the phase change material, the phase change material may be changed from solid phase to liquid phase by absorbing heat energy E of the system. When the phase change material melts from solid phase to liquid phase, the phase change material may absorb heat energy E corresponding to melting enthalpy.

[0092] According to an embodiment, as the temperature of the battery **189** increases, the phase change material may undergo a phase changing. For example, when a short circuit occurs due to contact between the first electrode **310** and the second electrode **320**, a short circuit current may flow inside the battery **189**. When heat is generated by the short circuit current, the temperature of the battery **189** may increase. When the temperature of the battery **189** increases, a phase change material in the core **332** may be changed from solid phase to liquid phase by absorbing heat energy E inside the battery **189**. The heat energy E, which is absorbed when the phase change material undergoes the phase change, may be stored as latent heat in the phase change material. Since the heat energy E may be stored as the latent heat in the phase change material, a temperature increase of the battery **189** may be reduced. Even in a case that a short circuit occurs, the battery **189** according to an embodiment may reduce the temperature increase by the phase change material included in the core **332** of the separator **330**. The fire of the battery **189** may be reduced by controlling a rapid temperature increase of the battery **189** through the phase change material.

[0093] For example, an electronic device (e.g., the electronic device **101** of FIG. **2B**) including the battery **189** may be subjected to an external impact. For example, the electronic device **101** may be dropped from a high position to a low position, or may collide with a physical external object O.

When the electronic device **101** is subjected to an external impact, the impact may be transmitted to the battery **189** included inside the electronic device **101**. The impact applied to the battery **189** may cause a short circuit inside the battery **189**. The short circuit inside the battery **189** may cause a temperature increase of the battery **189**, thereby causing the battery **189** to catch fire. When the battery **189** catches fire, other components within the electronic device **101** may be damaged, which may cause unexpected harm to the user. According to an embodiment, when the temperature of the battery **189** increases, a phase change material within the separator **330** may absorb heat energy E and undergo a phase changing, thereby lowering the temperature of the battery **189**. As the temperature increase of the battery **189** is suppressed, the occurrence of the battery **189** catching fire may be reduced.

[0094] According to an embodiment, in addition to the short circuit of the battery **189**, the temperature increase of the battery **189** due to various causes may be reduced. For example, while charging the battery **189**, the temperature of the battery **189** may increase. As the temperature of the battery **189** increases, the electrical conductivity decreases, which may deteriorate the charging performance. According to an embodiment, the phase change material in the separator **330** may reduce the temperature increase of the battery **189** by absorbing heat energy E. Since the temperature increase of the battery **189** may be suppressed during a charging process, the charging efficiency of the battery **189** may be improved.

[0095] According to an embodiment, the phase change material may include a material that is capable of repeatedly performing melting and solidification processes and having a high latent heat. For example, paraffin may be suitable as a phase change material because it has a relatively high latent heat, enables easy control of a phase change temperature depending on the molecular weight, and is inexpensive.

[0096] Referring to FIG. 6B, when the phase change material in a separator **330** absorbs thermal energy E of a battery **189**, the separator **330** may expand. For example, when the temperature of the battery **189** increases, a size of pores **330a** in the separator **330** may decrease. The separator **330** may have a shutdown function in which the pores **330a** are closed to reduce the thermal runaway phenomenon when the battery **189** is overheated. As the pores **330a** may be dissolved by the heat, the movement of ions may decrease. As the pores **330a** are closed, the movement of ions in the electrolyte **350** may decrease, but the first electrode **310** and the second electrode **320** may be in contact with each other due to a contraction of the separator **330**. When the first electrode **310** and the second electrode **320** are in contact with each other, an internal short circuit may occur, causing the temperature of the battery **189** to rapidly increase.

[0097] According to an embodiment, as the temperature of the battery **189** increases, the phase change material within the core **332** may be changed from solid phase to liquid phase. As the phase change material is changed to liquid phase, a volume of the phase change material may increase. For example, as the phase change material within the core **332** is changed to liquid phase, a volume of the core **332** may increase by about 10%, but is not limited thereto.

[0098] According to an embodiment, when the phase change material included in the core **332** is changed from solid phase to liquid phase, the phase change material in the liquid phase may be surrounded by the sheath **333**. According to an embodiment, a melting point of a polymer material may be higher than a melting point of the phase change material. Even when the phase change material undergoes the phase change, the polymer material does not melt, so the sheath **333** may surround at least a portion of the core **332**. The sheath **333** may elongate based on the phase change of the phase change material. Since the sheath **333** surrounds at least a portion of the core **332**, the sheath **333** may expand by an increase in a volume of the core **332**. The sheath **333** may elongate while maintaining a core-sheath structure. As the volume of the core **332** increases, the fibers **331** forming the separator **330** may expand as the sheath **333** surrounding the core **332** elongates. As the fibers **331** forming the separator **330** expand, the separator **330** may expand. As the separator **330** expands, the separator **330** may compensate for the contraction of the separator **330** due to the

temperature increase. The polymer material included in the sheath **333** may include a material capable of providing mechanical strength so that the sheath **333** is not damaged when elongated. For example, the polymer material may include metallic oxide, but is not limited thereto.

[0099] For example, when an internal short circuit occurs due to the contact between the first electrode **310** and the second electrode **320**, the temperature of the battery **189** may increase. The phase change material in the core **332** of the separator **330** may be changed from solid phase to liquid phase by absorbing heat energy E. As the phase change material undergoes the phase change, a volume of the core **332** may increase and the sheath **333** surrounding the core **332** may elongate, thereby increasing a width of the fiber **331**. As the width of the fiber **331** increases, the separator **330** may expand. The expansion of the separator **330** may compensate for a contraction of the separator **330** due to an increase in temperature to reduce the contraction of the separator **330**. According to an embodiment, as the contraction of the separator **330** is compensated, the separator **330** may physically separate the first electrode **310** and the second electrode **320** and prevent the internal short circuit. The battery **189** according to an example may suppress a temperature increase of the battery **189** and prevent a short circuit by compensating for the contraction of the separator **330**. The battery **189** according to an example may provide thermal stability.

[0100] FIG. 7 illustrates a state in which an external object penetrates a battery according to an embodiment of the disclosure.

[0101] Referring to FIG. 7, a battery **189** may be damaged by an external object O. For example, the external object O (e.g., a nail or a screw), which is rigid and conductive, may penetrate the battery **189**. The external object O may electrically connect a first electrode **310** and a second electrode **320** by penetrating from the first electrode **310** to the second electrode **320**. For example, when the external object O includes a conductive material (e.g., metal), the first electrode **310** and the second electrode **320** may be electrically connected through the external object O penetrating the battery **189**. When the first electrode **310** and the second electrode **320** are electrically connected, a short circuit may be formed in the battery **189**. When the short circuit is formed, a short circuit current may flow along the first electrode **310**, the external object O, and the second electrode **320**. The short circuit current may generate heat. Due to the generated heat, a temperature of a damaged portion of the battery **189** may increase rapidly, which may cause the battery **189** to catch fire.

[0102] According to an embodiment, when the external object O penetrates the battery **189**, the external object O may penetrate a separator **330**. The short circuit of the battery **189** may occur due to the external object O, and the short circuit may cause a temperature increase of the battery **189**. As the temperature of the battery **189** increases, a phase change material in a core **332** may be changed from solid phase to liquid phase by absorbing heat energy. Since the phase change material absorbs heat energy, the temperature increase of the battery **189** due to the short circuit may be reduced.

[0103] According to an embodiment, when the external object O penetrates the battery **189**, the phase change material in the core **332** may surround the external object O penetrating the separator **330**. Referring to FIG. 7, the external object O (e.g., a nail) may be inserted from the first electrode **310** to the second electrode **320** by penetrating the separator **330**. When the external object O penetrates the separator **330**, a core-sheath structure at a penetration point may be damaged. Since a sheath **333** at the penetration point is damaged, the phase change material in the core **332** may move to the outside of the separator **330**. The phase change material may move around the external object O inserted into the second electrode **320** at the penetration point, thereby surrounding the external object O. In a state of surrounding the external object O, the phase change material may be changed from solid phase to liquid phase by absorbing heat energy. Since the phase change material may absorb heat energy in a state of surrounding the external object O, formation of a short-circuit current may be suppressed. For example, the short circuit current may be formed from the first electrode **310** to the second electrode **320** along the external object O. As the phase change

material surrounds the external object O at the penetration point, a flow of the short-circuit current flowing from the first electrode **310** to the second electrode **320** through the external object O may be suppressed. According to an embodiment, even when the external object O penetrates the battery **189**, the phase change material in the separator **330** may suppress a temperature increase of the battery **189** and a flow of the short-circuit current. According to an embodiment, since the battery **189** catching fire may be suppressed, stability of the battery **189** may be secured.

[0104] A battery (e.g., the battery **189** of FIG. 3C) according to an embodiment may include a first electrode (e.g., the first electrode **310** of FIG. 3C), a second electrode (e.g., the second electrode **320** of FIG. 3C), and a separator (e.g., the separator **330** of FIG. 3C). The second electrode may be spaced apart from the first electrode. The separator may be disposed between the first electrode and the second electrode. The separator may include a fiber (e.g., the fiber **331** of FIG. 4B). The fiber may include a core (e.g., the core **332** of FIG. 5A) and a sheath (e.g., the sheath **333** of FIG. 5A). The sheath may at least partially surround the core. The core may include a phase change material. The sheath may include a polymer material. According to an embodiment of the disclosure, the separator may be configured to reduce a temperature increase of the battery. For example, when the temperature of the battery increases due to an internal short circuit, the phase change material in the core may be changed from solid phase to liquid phase by absorbing heat energy. As the temperature increase of the battery is reduced by the phase change material, overheating and/or a fire in the battery may be reduced. The battery according to an embodiment may provide stability during use and efficiently supply power because it does not overheat easily. According to an embodiment, since the separator does not include a separate coating layer and may include a phase change material through a core-sheath structure, the separator may be relatively thin. Since the separator is thin, the battery may have a relatively high energy density.

[0105] According to an embodiment, the phase change material may be configured to reduce, when the temperature of the battery increases, the temperature increase of the battery by a phase changing from solid phase to liquid phase. According to an embodiment of the disclosure, when the phase change material is changed to liquid phase, the phase change material may absorb heat energy corresponding to melting enthalpy. Since the temperature increase of the battery may be reduced by the phase change material, stability may be ensured when the battery is used.

[0106] According to an embodiment, as the phase change material in the core is changed from solid phase to liquid phase when the temperature of the battery increases, the separator may expand. The separator may compensate for a contraction of the separator through the expansion of the separator. According to an embodiment of the disclosure, when the temperature of the battery increases, pores in the separator may be closed. As the pores are closed, the separator may contract. The contraction of the separator may cause a short circuit in the battery. According to an embodiment, a volume of a phase change material in the core may increase when changing from solid phase to liquid phase. Through an increase in the volume of the phase change material, the fiber including the core and the sheath may expand. Since the fiber forming the separator expands, the separator may expand. The expansion of the separator may compensate for the contraction of the separator due to the temperature increase of the battery. According to an embodiment, since the separator may compensate for the contraction of a portion that contracts when the temperature of the battery increases, a short circuit due to contact between a positive electrode and a negative electrode may be prevented.

[0107] According to an embodiment, the sheath may be elongated based on the phase changing of the phase change material. According to an embodiment of the disclosure, the sheath may be elongated based on the expansion of the core. Since the sheath surrounds at least a portion of the core, it is elongated according to the expansion of the core, thereby reducing movement of the polymer material in the core to the outside of the fiber.

[0108] According to an embodiment, a melting point of the polymer material may be higher than a melting point of the phase change material. According to an embodiment of the disclosure, since

the melting point of the polymer material in the sheath is higher than the melting point of the phase change material in the core, even when the phase change material is melted, it may maintain a solid phase state. By maintaining the solid phase state of the polymer material, the polymer material may maintain the core-sheath structure.

[0109] According to an embodiment, the phase change material may include at least one of a paraffin, polyethylene glycol, sodium acetate trihydrate, sodium hydroxide monohydrate, magnesium nitrate hexahydrate, myristic acid, stearic acid, and xylitol. According to an embodiment of the disclosure, the phase change material may include a material having a relatively high latent heat and capable of repeatedly performing a melting and solidification process. Since the high latent heat of the phase change material may absorb and store a relatively large amount of heat energy, the separator may suppress the temperature increase of the battery.

[0110] According to an embodiment, the polymer material may include at least one of polyvinylidene fluoride (PVDF), polydimethylsiloxane (PDMS), polyethylene (PE), polyvinyl alcohol (PVA), and polyvinyl pyrrolidone (PVP).

[0111] According to an embodiment, the polymer material may include metallic oxide. According to an embodiment of the disclosure, the polymer material in the sheath may include a material that provides stability of the separator by providing mechanical strength of the separator.

[0112] According to an embodiment, a melting point of the phase change material may be 28° C. to 90° C. According to an embodiment of the disclosure, the phase change material may be melted at a temperature higher than the melting point. When the phase change material is melted, the temperature increase of the battery may be reduced by absorbing heat energy.

[0113] According to an embodiment, a porosity of the separator may be 30 to 80%. According to an embodiment of the disclosure, the separator may include pores for enabling passage of ions. Since ions (e.g., lithium cations) may move through the pores, charging and/or discharging of the battery may be possible.

[0114] According to an embodiment, a width (e.g., the width w2 of FIG. 5A) of the core may be greater than or equal to 50% of a width (e.g., the width w1 of FIG. 5A) of the fiber.

[0115] According to an embodiment, a width of the fiber may be from 1 nm to 1,000 nm.

According to an embodiment of the disclosure, the width of the fiber may vary according to the type, size, and use of the battery.

[0116] According to an embodiment, the core may include 10 weight percent (wt %) to 90 wt % of the phase change material.

[0117] According to an embodiment, the battery may further include a case (e.g., the case 340 of FIG. 3A) for accommodating the first electrode, the second electrode, and the separator. The sheath may be in contact with an electrolyte (e.g., the electrolyte 350 of FIG. 3A) in the case. According to an embodiment of the disclosure, the separator may include a phase change material in the core without including a coating layer formed of a phase change material. The separator including a fiber of a core-sheath structure may be relatively thin because it does not include a separate coating layer. According to an embodiment, the battery may have a relatively high energy density.

[0118] According to an embodiment, the phase change material may be configured to reduce, when an external object (e.g., the external object O of FIG. 7) penetrates the separator, a temperature increase of the battery by surrounding the external object. According to an embodiment of the disclosure, the phase change material may surround, when the external object penetrates the battery, the external object at a penetration portion. When the external object penetrates the battery, a short circuit may be formed inside the battery. Due to the short circuit, the temperature of the battery may increase, and the phase change material may be changed from solid phase to liquid phase by absorbing heat energy. The phase change material in the liquid phase may block a flow of a short circuit current formed through the external object by surrounding the external object.

According to an embodiment, in a state that the external object penetrates the battery, the separator may suppress the temperature increase of the battery and reduce ignition.

[0119] An electronic device (e.g., the electronic device **101** of FIG. 2B) according to an embodiment may include a battery (e.g., the battery **189** of FIG. 3A) and PMIC (e.g., the PMIC **360** of FIG. 2B). The battery may be configured to provide power to at least one component of the electronic device. The PMIC may be configured to manage power provided from the battery to the at least one component. The battery may include a first electrode, a second electrode, and a separator. The second electrode may be spaced apart from the first electrode. The separator may be disposed between the first electrode and the second electrode. The separator may include a fiber. The fiber may include a core and a sheath. The sheath may at least partially surround the core. The core may include a phase change material. The sheath may include a polymer material. According to an embodiment of the disclosure, the separator may be configured to reduce a temperature increase of the battery. For example, when the temperature of the battery increases due to an internal short circuit, the phase change material in the core may be changed from solid phase to liquid phase by absorbing heat energy. As the temperature increase of the battery is reduced by the phase change material, overheating and/or ignition of the battery may be reduced. Since the battery according to an embodiment does not easily overheat, it may provide stability during use and efficiently provide power. According to an embodiment, since the separator may include a phase change material through a core-sheath structure without including a separate coating layer, a thickness of the separator may be relatively thin. Since the thickness of the separator is thin, the battery may have a relatively high energy density.

[0120] According to an embodiment, the phase change material may include at least one of a paraffin, polyethylene glycol, sodium acetate trihydrate, sodium hydroxide monohydrate, magnesium nitrate hexahydrate, myristic acid, stearic acid, and xylitol. According to an embodiment of the disclosure, the phase change material may include a material having a relatively high latent heat and capable of repeatedly performing a melting and solidification process. Since the high latent heat of the phase change material may absorb and store a relatively large amount of heat energy, the separator may suppress the temperature increase of the battery.

[0121] According to an embodiment, the polymer material may include at least one of polyvinylidene fluoride (PVDF), polydimethylsiloxane (PDMS), polyethylene (PE), polyvinyl alcohol (PVA), and polyvinyl pyrrolidone (PVP).

[0122] According to an embodiment, the polymer material may include metallic oxide. According to an embodiment of the disclosure, the polymer material in the sheath may include a material capable of providing stability of the separator by providing mechanical strength of the separator.

[0123] According to an embodiment, a melting point of the phase change material may be 28° C. to 90° C. According to an embodiment of the disclosure, the phase change material may be melted at a temperature higher than the melting point. When the phase change material is melted, the temperature increase of the battery may be reduced by absorbing heat energy.

[0124] The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

[0125] It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of or all possible combinations of the items enumerated together in a

corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” or “connected with” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0126] As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0127] Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between a case in which data is semi-permanently stored in the storage medium and a case in which the data is temporarily stored in the storage medium.

[0128] According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0129] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0130] It will be appreciated that various embodiments of the disclosure according to the claims and description in the specification can be realized in the form of hardware, software or a combination of hardware and software.

[0131] Any such software may be stored in non-transitory computer readable storage media. The non-transitory computer readable storage media store one or more computer programs (software modules), the one or more computer programs include computer-executable instructions that, when executed by one or more processors of an electronic device individually or collectively, cause the electronic device to perform a method of the disclosure.

[0132] Any such software may be stored in the form of volatile or non-volatile storage such as, for example, a storage device like read only memory (ROM), whether erasable or rewritable or not, or in the form of memory such as, for example, random access memory (RAM), memory chips, device or integrated circuits or on an optically or magnetically readable medium such as, for example, a compact disk (CD), digital versatile disc (DVD), magnetic disk or magnetic tape or the like. It will be appreciated that the storage devices and storage media are various embodiments of non-transitory machine-readable storage that are suitable for storing a computer program or computer programs comprising instructions that, when executed, implement various embodiments of the disclosure. Accordingly, various embodiments provide a program comprising code for implementing apparatus or a method as claimed in any one of the claims of this specification and a non-transitory machine-readable storage storing such a program.

[0133] While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

Claims

1. A battery comprising: a first electrode; a second electrode spaced apart from the first electrode; and a separator including a fiber disposed between the first electrode and the second electrode, the fiber including a core and a sheath at least partially surrounding the core, wherein the core of the fiber includes a phase change material, and wherein the sheath of the fiber includes a polymer material.
2. The battery of claim 1, wherein, when a temperature of the battery increases, the phase change material is configured to reduce a temperature increase of the battery by a phase changing from solid phase to liquid phase.
3. The battery of claim 1, wherein, when a temperature of the battery increases, the separator is configured to expand as the phase change material in the core changes from solid phase to liquid phase, and wherein a contraction of the separator is compensated through the expansion of the separator.
4. The battery of claim 1, wherein the sheath of the fiber is configured to elongate based on a phase changing of the phase change material.
5. The battery of claim 1, wherein a melting point of the polymer material is higher than a melting point of the phase change material.
6. The battery of claim 1, wherein the phase change material includes at least one of a paraffin, polyethylene glycol, sodium acetate trihydrate, sodium hydroxide monohydrate, magnesium nitrate hexahydrate, myristic acid, stearic acid, or xylitol.
7. The battery of claim 1, wherein the polymer material includes at least one of polyvinylidene fluoride (PVDF), polydimethylsiloxane (PDMS), polyethylene (PE), polyvinyl alcohol (PVA), and polyvinyl pyrrolidone (PVP).
8. The battery of claim 1, wherein the polymer material includes a metallic oxide.
9. The battery of claim 1, wherein a melting point of the phase change material is from 28° C. to 90° C.
10. The battery of claim 1, wherein a porosity of the separator is from 30% to 80%.
11. The battery of claim 1, wherein a width of the core is 50% or more of a width of the fiber.

- 12.** The battery of claim 1, wherein a width of the fiber is from 1 nanometer (nm) to 1,000 nm.
 - 13.** The battery of claim 1, wherein the phase change material is from 10 weight percent (wt %) to 90 wt % of the core of the fiber.
 - 14.** The battery of claim 1, further comprising: a case configured to accommodate the first electrode, the second electrode, and the separator, wherein the sheath of the fiber contacts an electrolyte in the case.
 - 15.** The battery of claim 1, wherein, when an external object penetrates the separator, the phase change material is configured to reduce a temperature increase of the battery by surrounding the external object.
 - 16.** An electronic device comprising: a battery configured to provide power to at least one component of the electronic device; and power management integrated circuitry configured to manage the power provided from the battery to the at least one component, wherein the battery comprises: a first electrode, a second electrode spaced apart from the first electrode, and a separator including a fiber disposed between the first electrode and the second electrode, the fiber including a core and a sheath at least partially surrounding the core, wherein the core of the fiber includes a phase change material, and wherein the sheath of the fiber includes a polymer material.
 - 17.** The electronic device of claim 16, wherein the phase change material includes at least one of a paraffin, polyethylene glycol, sodium acetate trihydrate, sodium hydroxide monohydrate, magnesium nitrate hexahydrate, myristic acid, stearic acid, or xylitol.
 - 18.** The electronic device of claim 16, wherein the polymer material includes at least one of polyvinylidene fluoride (PVDF), polydimethylsiloxane (PDMS), polyethylene (PE), polyvinyl alcohol (PVA), or polyvinyl pyrrolidone (PVP).
 - 19.** The electronic device of claim 16, wherein the polymer material includes a metallic oxide.
 - 20.** The electronic device of claim 16, wherein a melting point of the phase change material is from 28° C. to 90° C.
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